LIAM McBRIDE

Data Scientist · Statistical Modeling · Machine Learning · High-Dimensional Data

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EDUCATION

M.Sc. Bioinformatics for Health Sciences Oct 2024 - June 2026

Universitat Pompeu Fabra (UPF) — Barcelona, Spain

Relevant coursework: Statistical & Machine Learning Methods, Databases & Web Development (SQL), Algorithms, Computational Genomics

B.A. Biology Aug 2019 – May 2024

St. Olaf College — Northfield, MN

Concentrations: Mathematical Biology; Statistics & Data Science · Coursework: Probability Theory, Statistical Modeling I & II, Algorithms for Decision Making, Intro to Data Science, Modern Computational Mathematics · Presidential Scholarship, Dean's List (2021–2022)

TECHNICAL SKILLS

Languages: Python, R, SQL (familiar), Bash, Git

Data Science & ML: scikit-learn, PyTorch, pandas, NumPy, SciPy, statsmodels, tidyverse

Visualization & Reporting: ggplot2, matplotlib, Seaborn, Shiny, RMarkdown, Jupyter

Data Engineering: Docker, Nextflow (familiar), HPC/cluster computing, BioPython

Domain Expertise: High-dimensional genomics data, statistical hypothesis testing, pipeline development, reproducible research

EXPERIENCE

Bioinformatics Research Intern (Master's Thesis) · *IMIM – Hospital del Mar Research Institute / UPF* Oct 2025 – Present

Barcelona, Spain

- Designed and implemented end-to-end data pipelines in R and Python to process, normalize, and analyze high-dimensional single-cell RNA-seq datasets (tens of thousands of features per sample) from an Alzheimer's disease mouse model
- Applied unsupervised clustering, dimensionality reduction (PCA, UMAP), and statistical modeling to identify cell-type-specific gene expression patterns across experimental conditions and biological sexes
- Conducted differential expression analysis and pathway enrichment using curated gene sets (MitoCarta) to extract biologically meaningful signals from complex, noisy data
- Integrated multiple data modalities — transcriptomic profiles, gene regulatory networks, and behavioral phenotype data — to build a multi-layered view of treatment response

Bioinformatics Intern · *Dana-Farber Cancer Institute, Harvard University* May 2023 – Jan 2024

Boston, MA

- Built and validated a data processing pipeline combining sequencing outputs with statistical clonality assessment methods for lymphoma diagnostics from cell-free DNA samples
- Designed and executed a systematic benchmarking study comparing multiple analytical pipelines; synthesized quantitative results into a comparative report to support data-driven platform selection

Computational Biology Intern · *University of Minnesota Medical School* Jun – Aug 2022

Minneapolis, MN

- Processed and analyzed whole-genome sequencing data using Python-based pipelines to investigate genetic elements associated with a pediatric cancer; managed large-scale genomic data from ingestion through interpretation
- Performed statistical analysis and built visualizations in R to communicate findings from complex bioinformatic pipelines to a multidisciplinary research team

Research Assistant · *Smith College* Mar 2017 – May 2021

Northampton, MA (including COVID-period gap year, 2020–2021)

- Developed a Python-based data extraction pipeline to process large collections of biological sequence files from public databases, enabling downstream phylogenetic analysis

- Applied statistical tools and databases to analyze genome-scale datasets and reconstruct evolutionary relationships across species

Student Researcher · *St. Olaf College, Biology Department*Jun – Aug 2021
Northfield, MN

- Designed an experiment and performed statistical analysis and data visualization in R to study environmental nutrient dynamics; presented findings at a college-wide research symposium

TEACHING

R Tutor · *St. Olaf College*2022 – 2024
Northfield, MN

- Supported students in biology and statistics courses with R programming, data wrangling, visualization, and reproducible analysis workflows in RStudio and RMarkdown

PUBLICATIONS

- Antel K., Xiong N., Merrill M., Chen T., McGrath L., Droessler C., van der Schyff M., Wu M.X., Cheddie D., Slome C., McBride L., Day J., Mohamed Z., Fakie N., Mbatani N., Shultz C., Meintjes G., Tyekucheva S., Cohen E.F., Tolstorukov M.Y., Verburgh E., Murakami M.A. (2026). Access to diagnosis using liquid biopsy (ADLiB): Identifying lymphoma in a tuberculosis-endemic setting. HemaSphere, 10(6), e70394. https://doi.org/10.1002/hem3.70394

PRESENTATIONS & POSTERS

- McBride L., Ramon-Duaso C., Mato Blanco X., Berenguer-Molins P., Perera J., Busquets-Garcia A. (2026). Computational Tools to Link Cannabinoids to Alzheimer's Disease. Innovations in Single Cell Omics (ISCO), IMIM, Barcelona.
- McBride L., Xiong N., Antel K., Murakami M. (2024). Quantifying Clonotype Diversity within T-cell and B-cell Acute-Lymphoblastic-Leukemia Patients. St. Olaf Mathematical Biology Symposium.
- Dix M., Wang K., McBride L., Freedberg S. (2022). The Role of Species-Level Competition in the Evolution of Altruism and Selfishness. Honors Day, St. Olaf College.